

NRES 746, Week 6, Lecture 2: Find the Bugs

Name: _____

Instructions: Work alone for about 3 minutes, then compare with a neighbor. Keep laptops closed.

The code

A classmate wrote these two functions for last week's Lab 4, Act 2 (the toad monitoring power analysis), possibly with help from an AI assistant. As a reminder:

- each animal is detected on one person-day with probability `pobs`, so a bout of `k` person-days detects it with probability $1 - (1 - p_{obs})^k$;
- `estimate_power()` should simulate `R` replicate surveys and return the fraction in which a decline was detected. Assume the functions it calls, `simulate_counts()` and `detect_decline()`, work correctly.

Each function below contains **exactly one bug**. Both run without an error, but give wrong answers.

```
detect_prob <- function(k, pobs = 0.02) {  
  1 - pobs^k  
}  
  
estimate_power <- function(k, s, R = 1000) {  
  t_survey <- seq(1, 25, by = s)  
  detected <- logical(R)  
  for (i in seq_len(R)) {  
    counts <- simulate_counts(t_survey, k)  
    detected <- detect_decline(t_survey, counts)  
  }  
  mean(detected)  
}
```

Part 1: Find and fix (individual, ~3 min)

For each function, circle the bug on the code above, write the corrected line, and say what would go wrong if you ran the buggy version.

1a. `detect_prob()`

1b. `estimate_power()`

Part 2: Catch it next time (with a neighbor, ~2 min)

2. For one of the bugs, what quick check (one line of code and the answer you'd expect) would have caught it?
